THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 2: **Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Water flowing in the outer jacket of a condenser flows from the bottom to the top because: 1

(a) to save water
(b) it is easier
(c) it keeps the jacket full for efficient cooling
(d) to avoid noise

2. Assertion (A): A saturated solution at a higher temperature deposits crystals on cooling. Reason (R): The solubility of most solids decreases as the temperature falls. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Dilution of a 12 per cent solution to 3 per cent requires the volume to be: 1

(a) increased four times **(b)** tripled
(c) decreased **(d)** doubled

4. Why are boiling chips added to the flask in distillation? 1

(a) to increase temperature
(b) to dissolve
(c) to clean
(d) to prevent bumping and give smooth boiling

5. Assertion (A): Colloidal particles pass through a filter paper. Reason (R): The size of colloidal particles is smaller than the pores of the filter paper. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A colloid in which a liquid is dispersed in a liquid is called: 1

(a) a sol **(b)** an aerosol **(c)** an emulsion **(d)** a foam

7. A liquid mixture of three components with boiling points 60, 80 and 120 degrees Celsius is separated by fractional distillation. The fraction collected at 80 degrees Celsius is: 1

(a) the first **(b)** a mixture of all three
(c) the second component **(d)** the third

8. Assertion (A): Distilled water is a pure substance. Reason (R): It has fixed composition and fixed boiling point. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** Paper chromatography can separate the different dyes present in black ink. 1
Reason (R): Different dyes have different solubilities in the solvent and are carried to different heights on the paper.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Air is a homogeneous mixture of gases. 1
Reason (R): Nitrogen forms about 78 per cent of air by volume.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is a mixture of sulphur and iron heated differently to separate than by a magnet? 2

12. Why does paper chromatography not work well if the spot is immersed in the solvent? 2

13. State why vinegar is a true solution and soap solution is a colloid. 2

14. Why is the addition of alum used in purifying muddy water? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the principles of fractional distillation and crystallisation are used in industry for sugar and for oxygen. 3

16. A mixture of water and alcohol is heated slowly. The temperature rises from 78 to 100 degrees Celsius. Interpret. 3

17. Explain how the solubility curve is used to decide the mass of crystals and the choice of a solvent for crystallisation. 3

OR

Explain why a decanting technique is faster after adding alum to muddy water. 3

18. Explain why alloys like steel are harder than the pure metal iron. 3

OR

Explain the role of surface area and temperature in the rate of dissolving a solid and design a fair test. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A dairy in Anand separates cream from milk using a machine that spins the milk at high speed. The lighter cream collects near the centre while the heavier skimmed milk moves to the outside. 5
(i) Name the separation method used.
(ii) Which component of milk is lighter?
(iii) Give one other use of this method.
(iv) Explain the principle on which the method works.

20. Explain the preparation of a saturated solution at a given temperature, how to check that it is saturated and how to find the solubility of the salt at that temperature by an experiment. 5

OR

Discuss how the knowledge of mixtures and separation is important in medicines, food, mining and environmental protection with examples. 5

END OF QUESTION PAPER — Total: 40 Marks